

Integrating Computer Vision : Agent-Based Crowd Simulation for Stampede Prevention and Event Safety

Project Review - I

Project Team:

Nivin P Louis (TL23BTAI0042)
Niranjana Sreevalsan (TL23BTAI0348)
Farhan Mohamed T P (TL23BTAI0308)
Nihal P Shakeer (TL23BTAI0249)

Project Guide:

Anju K P, Assistant Professor, Dept. AIML

Vidya Academy of Science and Technology, Thrissur



College Vision

To be a premier institution in engineering education and research, nurturing innovative professionals for sustainable societal development.

College Mission

M1: Provide quality engineering education that empowers students with strong technical competence and analytical problem solving abilities.

M2: Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable technological solutions.

M3: Develop ethical and socially responsible professionals with leadership qualities, lifelong learning abilities and a commitment to societal well-being.

Department Vision

To be a center of excellence in artificial intelligence and machine learning education, research, and innovation, nurturing competent professionals capable of developing intelligent, ethical, and sustainable solutions for societal and industrial advancement.

Department Mission

M1: Provide quality education in artificial intelligence and machine learning through effective teaching-learning practices, enabling students to acquire strong technical knowledge, analytical skills, and problem-solving abilities.

M2: Promote research, innovation, entrepreneurship, and interdisciplinary collaboration in emerging AI technologies to address industrial and societal challenges.

M3: Develop professionally competent graduates with ethical values, leadership qualities, lifelong learning abilities, and a commitment to societal well-being in a technology-driven world.

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Problem Statement

Problem Statement

Large public gatherings such as concerts, festivals, religious events, stadiums, and transport hubs are vulnerable to overcrowding, bottlenecks, panic movement, and stampede-like situations. Existing crowd-control practices are often manual and reactive, detecting danger only after congestion has already become severe.

- Traditional capacity checks do not capture dynamic crowd movement.
- CCTV monitoring alone identifies the present state but does not predict future risk.
- A proactive system is required to detect density, simulate crowd evolution, and alert event operators before critical crowd pressure develops.

Objectives

- **Develop a CV Module:** Detect people from live camera/video feeds and estimate real-time crowd count and zone-wise density.
- **Develop an Agent-Based Model (ABM):** Model each person as an autonomous agent with goals, local perception, and collision avoidance rules.
- **Simulate Realistic Crowd Psychology:** Include panic propagation and agent pressure to model transition from calm movement to risky crowd behavior.
- **Model Complex Environments:** Represent exits, corridors, obstacles, one-way paths, narrow gates, and bottleneck regions.
- **Visualize Risk Zones:** Generate density heatmaps, congestion maps, and risk alerts for event safety management.
- **Evaluate Intervention Strategies:** Compare safety impact of rerouting, exit optimization, and guided crowd-control protocols.

Motivation

- **Critical Safety Risks:** Large-scale gatherings are increasingly prone to dangerous overcrowding and stampedes, often caused by inadequate infrastructure planning rather than the crowd itself.
- **Limitations of Traditional Models:** Standard "fluid-based" models treat crowds as a homogeneous mass, failing to capture the complex, individual decision-making (micro-interactions) that leads to disasters.
- **Emergent Behavior Complexity:** Dangerous phenomena like "contagious panic" and crushing pressures arise from individual interactions that are difficult to predict without granular simulation.
- **Cost of Trial and Error:** Testing crowd control measures in real-world scenarios is dangerous and impossible; a virtual environment is required to safely test "what-if" emergency scenarios.
- **Need for Dynamic Layout Analysis:** Static capacity calculations cannot account for dynamic bottlenecks (e.g., security checks, narrow gates) where flow rates fluctuate rapidly.

Literature Review — Paper 1

Haoyang Liang, Jian Sun et al. “Toward Interactive Crowd Dynamics in Virtual Built Environment: A BIM-Driven and Macro-Micro Integrated Crowd Simulation Framework,” *Developments in the Built Environment* (Elsevier), 2026.

Objective

- Enhance crowd simulation fidelity in virtual environments by integrating BIM data with agent movement models.

Methodology

- **Extract:** Derive semantic mesh graphs from 3D BIM data.
- **Formulate:** Compute global routing fields via a potential field model.
- **Simulate:** Execute local agent adaptations using a social force framework.

Models Used

BIM Navigation Graph | Potential Field Model

Component	Model/Value
HIL Participants	28
Baseline Route	A-Star
Peak Density	5 ped/m ²
Stress Capacity	5000 Agents

Research Gap

- No support for physical pushing under extreme panic.
- Severe frame rate drop at 5000 agents.
- Velocity delay triggers occasional late collisions.

Literature Review — Paper 2

Geetanjali Bhola, Sumit Srivastava et al. “Stampede Detection and Crowd Analysis Using CNN-LSTM and Farneback Optical Flow,” *Scientific Reports* (Springer), 2026.

Objective

- Classify crowd risk into 4 levels (normal, moderate, dense, risky) and detect stampedes.

Methodology

- **Motion:** Extract Farneback dense optical flow features.
- **Modeling:** Train CNN-LSTM on 10k frames for spatiotemporal dynamics.
- **Validation:** Run domain transfer tests on the UMN dataset.

Models Used

Farneback Optical Flow — CNN-LSTM

Metric	Value
Train Acc.	99.75%
Val. Acc.	99.23%
Throughput Window	12.77 FPS 16 Frames

Research Gap

- Performance drops under cross-dataset domain shifts.
- Dense optical flow is computationally expensive.
- Background motion triggers false alarms.
- Short sequence window misses long-term risk.

Literature Review — Paper 3

Wahida Mansouri, Manal Abdullah Alohalı et al. “Deep Convolutional Neural Network-Based Enhanced Crowd Density Monitoring for Intelligent Urban Planning on Smart Cities,” *Scientific Reports* (Springer), 2025.

Objective

- Classify crowd density under varying visual conditions to support smart-city urban planning and public safety.

Methodology

- **Preprocessing:** Use Gaussian filtering for image noise reduction.
- **Learning:** Extract features via SE-DenseNet; tune parameters with Red Fox Optimization.
- **Classification:** Deploy ConvLSTM to recognize density categories.

Models Used

SE-DenseNet — ConvLSTM

Metric	Value
Overall Acc.	98.40%
Optimizer	Red Fox (RFO)
Feature Ext.	SE-DenseNet
Temporal Clf.	ConvLSTM

Research Gap

- Constrained by static, pre-defined population settings.
- Static learning rates limit optimization performance.
- Computationally unscalable on massive datasets.
- Lacks real-time spatial adaptive learning.

Literature Review — Paper 4

Sarwar Khan, Zhigang Deng “Agent-Based Crowd Simulation: An In-Depth Survey of Determining Factors for Heterogeneous Behaviour,” *The Visual Computer* (Springer), 2024.

Objective

- Review microscopic crowd simulation models to classify heterogeneous agent behaviors and map open research fields.

Methodology

- **Search:** Query Scopus, Google Scholar, IEEE, and ACM databases.
- **Filter:** Apply inclusion/exclusion filters for agent-based studies.
- **Taxonomy:** Group papers by locomotion, emotion, roles, and physiology.

Models Used

Systematic Review (PRISMA)
Factor Taxonomy

Metric / Scope	Value
Identified Papers	383
Selected Studies	107
Dominant Factor	Locomotion
Secondary Factor	Personality/Emotion

Research Gap

- Over-reliance on navigation mechanics over crowd psychology.
- Weak integration of empirical real-world data sources.
- Isolated individual behaviors without holistic factor synergy.

Literature Review — Paper 5

Fumiyasu Makinoshima, Yusuke Oishi “Crowd Flow Forecasting via Agent-Based Simulations with Sequential Latent Parameter Estimation from Aggregate Observation,” *Scientific Reports* (Springer), 2022.

Objective

- Forecast microscopic crowd flows in real time by assimilating aggregate density maps into agent models.

Methodology

- Simulate:** Run force-based agent evacuations.
- Estimate:** Use particle filters for latent parameters.
- Analyze:** Optimize camera field-of-view placement.

Models Used

Force-Based Model — Particle Filter

Metric / Setup	Value
Simulated Agents	~5,000
Assimilation Type	Latent Parameter
Sensor Strategy	Optimal Placement
Scenarios	Evacuation

Research Gap

- Highly reliant on oversimplified pedestrian force rules.
- High computational cost scaling with massive crowds.
- Accuracy drops in complex architectural environments.

Literature Review — Paper 6

Omar Alidmat, Hamza Abu Owida et al. "Simulation of Crowd Evacuation in Asymmetrical Exit Layout Based on Improved Dynamic Parameters Model," *IEEE Access* (IEEE), 2025.

Objective

- Simulate multi-exit crowd evacuation in asymmetrical layouts using an improved cellular automaton.

Methodology

- **Discretize:** Divide room into cells using Moore neighborhood.
- **Calculate:** Compute transition payoffs via weighted DPM parameters.
- **Adapt:** Scale exit counting areas dynamically each time step.

Models Used

2D Cellular Automaton — Enhanced DPM (EDPM)

Metric / Setup	Value
Layout Scenario	20×20 Grid (4 Exits)
Movement Rules	Moore (8 Dir.)
Eval. Metrics	Evac Time, Dist.
Param. Weights	$\eta=0.5, \epsilon=0.15, \gamma=0.35$

Research Gap

- No real-world empirical data used for calibration.
- Random conflict resolution oversimplifies agent behavior.
- Fixed cell size ignores heterogeneous body dimensions.

Literature Review — Paper 7

Jinglin Xu, Yang Peng et al. “Hospital Flow Simulation and Space Layout Planning Based on Low-Trust Social Force Model,” *IEEE Access* (IEEE), 2024.

Objective

- Simulate hospital crowd behavior to reduce congestion and infection risks via a low-trust social force model.

Methodology

- **Formulate:** Develop LtSFM using variable mass for avoidance.
- **Simulate:** Integrate workflows with exponential infection decay models.
- **Evaluate:** Test layout against four spatial optimization schemes.

Models Used

Low-Trust Social Force Model — **Discrete Event Simulation**

Metric / Setup	Value
Simulation Runs	20 Repetitions
Data Window	6 Hours
Infection Rate	0.1%–1.0%
Layout Schemes	4 Planned

Research Gap

- Omits random staff walks and sign-following behaviors.
- Resource-intensive calibration for different hospital layouts.
- Assumes static infection dynamics without behavioral evolution.

Literature Review — Paper 8

Milad Malekzadeh, Klaus Bogenberger “The Future Walks With Us: How Robots Could Transform Pedestrian Dynamics and Urban Safety,” *IEEE Access* (IEEE), 2026.

Objective

- Assess the feasibility of integrating robots into pedestrian flow control and evacuation guidance scenarios.

Methodology

- **Review:** Search literature from 1995–2026 across major databases.
- **Categorize:** Classify robot roles into evacuation, flow, and guidance.
- **Evaluate:** Compare micro, macro, and hybrid crowd control models.

Models Used

Social Force Model — **Macroscopic PDEs**
— **Deep Reinforcement Learning**

Metric / Setup

Value

Timeframe	1995–2026
Databases	IEEE, Scopus, WoS
Domains	Evac, Flow, Guide
Model Scales	Micro, Macro, Hybrid

Research Gap

- No unified model for human-robot trust under panic.
- Lab simulations fail to generalize to real chaotic environments.
- Robotic systems expose cybersecurity and communication risks.

Literature Review — 9

W. Zhang, X. Xu “Distributed Event-Triggered Control for UAV Formation Tracking with Collision and Obstacle Avoidance,” *International Journal of Crowd Science*, 2026.

Objective

- Enable UAV swarms to track a maneuvering leader while reducing communication via event-triggered control.

Methodology

- **Estimate:** Design ESO and SMDO for leader state estimation.
- **Coordinate:** Build distributed controller with potential fields.
- **Optimize:** Apply event-triggered mechanism to cut communication.

Models Used

ESO — SMDO — APF — ETC
Framework

Metric / Setup	Value
Agents	4 Followers, 1 Leader
Sensing Range	10 m
Obstacle Range	3 m
Trigger Reduction	Up to 94.4%

Research Gap

- Relies on bounded disturbances and finite obstacle regions.
- No robustness analysis against cyber-attacks on network.
- Needs validation through physical UAV hardware experiments.

Literature Review — Paper 10

J. Xiong et al. “High-Dimensional Probability Modeling and Simulation for Restricted Crowd Walking Loads by Conditional Generative Adversarial Networks,” *Results in Engineering* (Elsevier), 2026.

Objective

- Model high-dimensional restricted crowd walking loads for structural vibration assessment using CGANs.

Methodology

- **Decompose:** Separate loads into pulse shapes and interval sequences.
- **Model:** Use FC-CGANs for waveforms and LSTM-CGANs for steps.
- **Validate:** Benchmark against force plate and footbridge data.

Models Used

Fully Connected CGAN — LSTM-based CGAN — SMD Model

Metric / Setup	Value
Guidance Frequencies	1.5, 1.8, 2.0 Hz
Pulse Vector Size	100 pts (1.0 s)
Validation Bridge	15 m Span
Test Subjects	12 Pedestrians

Research Gap

- Limited to three walking frequencies; restricts generality.
- Ignores human-structure interaction on flexible surfaces.
- Requires retraining for different layouts or walking paths.

Literature Review — Paper 11

Jiangbei Yue, Dinesh Manocha, He Wang “Human Trajectory Prediction via Neural Social Physics,” *Computer Vision – ECCV 2022, Lecture Notes in Computer Science* (Springer), 2022.

Objective

- Embed explicit motion physics inside deep learning models to improve trajectory accuracy and interpretability.

Methodology

- **Physics Core:** Apply learnable social-force equations.
- **Uncertainty:** Capture residual metrics via a VAE framework.
- **Testing:** Evaluate 15 deep models across 6 datasets.

Models Used

Neural Social Physics — **Stochastic VAE**

Metric / Setup

Value

Accuracy Gain	+5.56% to +70%
Density Transfer	2× to 5× Cap
Baselines Tested	15 Deep Models
Data Foundations	6 Standard Sets

Research Gap

- High formulation complexity impacts real-time frame rates.
- Remains constrained by pre-defined social force models.
- Predictability drops during sudden obstacle changes.

Literature Review — Paper 12

Khosro Rezaee et al. “A Survey on Deep Learning-Based Real-Time Crowd Anomaly Detection for Secure Distributed Video Surveillance,” *Personal and Ubiquitous Computing* (Springer), 2021.

Objective

- Review real-time anomaly detection in secure, distributed Web of Things (WoT) surveillance systems.

Methodology

- Taxonomy:** Map models (tracking, deep, hybrid).
- Metrics:** Evaluate latency, occlusion, and CPU load.
- Baseline:** Compare results using the MED dataset.

Models Used

Deep Networks — Hybrid WoT Models

Metric / Setup

Value

Review Focus	Real-Time Security
Reference Data	MED Dataset
Target Platform	WoT Edge Platforms
Top Performer	Hybrid Deep Learning

Research Gap

- Tracking-based models fail under heavy occlusions.
- Limited memory capacity on compact WoT edge nodes.
- Frequent false-positives in chaotic crowd patterns.

Literature Review — Paper 13

Doaa Mabrouk, Manal A. Abdel-Fattah, Ahmed Taha “Robust Crowd Anomaly Detection via Hybrid Ensemble Learning for Real-World Surveillance,” *Scientific Reports* (Springer), 2025.

Objective

- Deliver an interpretable crowd anomaly detection method that maintains high frame processing speeds on noisy or small datasets.

Methodology

- **Detection:** Locate crowd segments using YOLOv7 structures.
- **Features:** Extract optical-flow vectors for spatial motion shifts.
- **Ensemble:** Train Random Forest and Gradient Boosting blocks via Adam.

Models Used

YOLOv7 Network — RF + Gradient Boosting

Metric / Dataset	Value
UMN Accuracy	99.89%
Supermarket Acc.	92.60%
UMN Speed	24.0 FPS
Supermarket Speed	15.6 FPS

Research Gap

- Lacks spatial localization to pin down anomaly coordinates.
- Frame rates drop considerably on high-resolution streams.
- Optical-flow extraction remains sensitive to lighting changes.

Literature Review — Paper 14

Jun-Ho Choi, Jong-Hyeok Kim et al. “Three-Dimensional Atrous Inception Module for Crowd Behavior Classification,” *Scientific Reports* (Springer), 2024.

Objective

- Classify highly similar crowd behaviors in surveillance footage without causing parameter bloat.

Methodology

- Network:** Combine 3D atrous convolutions with inception blocks.
- Optimization:** Apply a custom Separation Loss for class imbalance.
- Evaluation:** Run benchmarks on the Crowd-11 behavior dataset.

Models Used

3D-AIM Framework — Separation Loss Engine

Metric / Setup	Value
Primary Dataset	Crowd-11
Target Loss Metric	Separation
Baseline Contrast	Binary CE
Stability Impact	Overfit Prevented

Research Gap

- Missing verification on real-world edge hardware systems.
- Computational demands scale quickly with 3D convolutions.
- Feature mapping accuracy drops during extreme occlusions.

Literature Review — Paper 15

Yang Liu, Dingkang Yang et al. “Generalized Video Anomaly Event Detection: Systematic Taxonomy and Comparison of Deep Models,” *ACM Computing Surveys* (ACM), 2024.

Objective

- Organize all anomaly detection settings into one unified, generalized framework.

Methodology

- **Taxonomy:** Group models by supervision and input modalities.
- **Resources:** Compile public datasets, codebases, and toolkits.
- **Analysis:** Quantitatively evaluate leading deep models.

Models Used

Deep VAD Architectures
GVAED Taxonomy

Metric / Setup	Value
Supervision Forms	4 Classes
Primary Focus	GVAED
Compiled Assets	Roadmap Data, Code, Metrics
Analysis Scope	Multi-Stream

Research Gap

- Evaluation metrics vary too widely across supervision limits.
- Inconsistent annotation quality limits weakly supervised models.
- Multi-stream networks introduce high resource footprints on edge.

Literature Review — Paper 16

Bingyu Zhao, Kelly Tang et al. “Dynamic Passenger Crowding and Operations in Urban Rail Transit Systems: A Validated Framework of Data-Driven Agent-Based Simulation,” *Journal of Rail Transport Planning & Management* (Elsevier), 2026.

Objective

- Develop a data-driven agent-based simulation to model dynamic passenger crowding in rail transit systems.

Methodology

- **Construct:** Build transit topology from GTFS and operational data.
- **Simulate:** Track passenger trips across 20-second time steps.
- **Calibrate:** Optimize parameters against smartcard tap-out data.

Models Used

Agent-Based Sim. (AAAM)
Shortest Dynamic Travel Time

Metric / Setup	Value
Case Study	Beijing Subway L6
Smartcard Data	2.3M Trips
Time Step	20 Seconds
Train Capacity	1960 Passengers

Research Gap

- Assumes uniform walking times, ignoring behavioral diversity.
- Passengers always board first train, oversimplifying choice.
- Sequential updates limit parallelization and scalability.

Literature Review — Paper 17

Kasereka K. Selain et al. “Enhancing Building Evacuation Efficiency in Case of Fire Breakout by Integrating Evacuee Categorization Into an Agent-Based Simulation,” *IEEE Access* (IEEE), 2025.

Objective

- Categorize evacuees by health and familiarity to optimize hospital fire evacuation routing.

Methodology

- **Categorization:** Group agents by health status and map familiarity.
- **Simulation:** Model dynamics inside a T-shaped hospital.
- **Benchmarking:** Test scenarios against literature baselines.

Models Used

Categorized ABM | Hospital Routing

Metric / Setup	Value
Hospital Layout	T-Shaped Geom.
Key Risk Zone	Central Intersect.
Categ. Basis	Health & Familiarity
Control Policy	Visit-Hour Caps

Research Gap

- Ignores real-time smoke toxicity and fire propagation rates.
- Behavior rules neglect caregiver-patient group bonding.
- Lacks hardware-in-the-loop dynamic routing trials.

Literature Review — Paper 18

Tarik Alafif, Mohammad Jassas et al. “Toward an Integrated Intelligent Framework for Crowd Control and Management (IICCM),” *IEEE Access* (IEEE), 2025.

Objective

- Unify AI, computer vision, and IoT into a centralized framework to manage safety in ultra-large crowds like the Hajj pilgrimage.

Methodology

- **Vision Layer:** Run density tracking and missing-person analytics.
- **IoT/Health Layer:** Use wearables and sensors for vitals mapping.
- **Control Layer:** Build a cloud decision engine for active routing.

Models Used

Computer Vision Stack
IoT Cloud Architecture

Metric / Setup	Value
Primary Case	Hajj Pilgrimage
Core Modalities	Vision, IoT, Health
Deployment	Large-Scale Ctrl.
Data Strategy	Cloud-Edge Sync

Research Gap

- Cloud latency bottlenecks real-time emergency notifications.
- Edge execution bounds are vulnerable to cyber threats.
- Lacks local, granular user privacy anonymization layers.

Literature Review — Paper 19

Ryo Niwa, Shunki Takami et al. “Estimate Crowd Flow Including Side-Trip Behavior When Exiting From Large-Scale Event Venues,” *IEEE Access* (IEEE), 2025.

Objective

- Model stadium side-trip exits using sparse observation data.

Methodology

- **Extension:** Add a side-trip parameter to shortest-path routing.
- **Data:** Calibrate parameters using Tokyo Dome LiDAR and 8K logs.
- **Testing:** Run cross-day generalizability benchmarks.

Models Used

**Shortest-Path + Side-Trip
Data Assimiler**

Metric / Setup	Value
Same-Day Acc.	+9.25%
Cross-Day Acc.	+3.15%
Test Environment	Tokyo Dome
Sensor Modality	LiDAR & 8K

Research Gap

- Single parameter simplifies complex commercial choice logic.
- Fails to reflect interior floorplan obstacles or blockages.
- High sensitivity to weather shifts affecting outdoor routes.

Literature Review — Paper 20

Almoaid A. Owaidah, Doina Olaru et al. “Modelling Mass Crowd Using Discrete Event Simulation: A Case Study of Integrated Tawaf and Sayee Rituals During Hajj,” *IEEE Access* (IEEE), 2021.

Objective

- Optimize scheduling and allocation for Tawaf and Sayee rituals to reduce mass congestion.

Methodology

- **Modeling:** Deploy a meso-level DES framework inside Grand Mosque.
- **Scenarios:** Simulate gate-closures and dynamic routing mix.
- **Metrics:** Evaluate overall completion counts and delay times.

Models Used

Meso-level DES | Allocation Optimizer

Metric / Setup	Value
Best Alloc. Range	30%–80%
Inefficient Strategy	Ground
Mitigation Design	Gate Closures
	Fitness Scheduling
Primary Site	Grand Mosque

Research Gap

- Meso-level DES models omit individual collision dynamics.
- Assumes steady group adherence to structural schedules.
- Neglects micro-level panic and sudden velocity drop-offs.

Project: Integrating Computer Vision & Agent-Based Crowd Simulation for Stampede Prevention

- 1 **Data Acquisition & Environment Setup** — Collect CCTV footage and public datasets (UCF-CC-50, ShanghaiTech, MOT); annotate head counts, bounding boxes, and flow vectors; model venue floor plans with exits, corridors, and choke points.
- 2 **Computer Vision Module** — Detect pedestrians per frame using **YOLOv8**; estimate crowd density per zone using **CSRNet**; track individuals with **DeepSORT** to extract velocity and direction fields.
- 3 **Agent-Based Model (ABM) Engine** — Spawn agents at positions derived from live CV density maps; apply the **Social Force Model (SFM)** for movement and collision avoidance; propagate panic states (calm $\rightarrow$ anxious $\rightarrow$ panic) via a threshold contagion model.
- 4 **Risk Analysis & Visualization** — Generate real-time density heatmaps on the venue floor plan; flag bottleneck zones (≥ 4 ped/m²) and compute crowd pressure index; trigger tiered alerts (Yellow / Orange / Red) and reroute exits via A* pathfinding.
- 5 **Validation & Evaluation** — Benchmark against CV-only and ABM-only baselines; evaluate using Precision, Recall, F1 (stampede detection) and ADE/FDE (trajectory); test across scenarios: concert exits, religious gatherings, and stadium evacuations.

Current Prototype

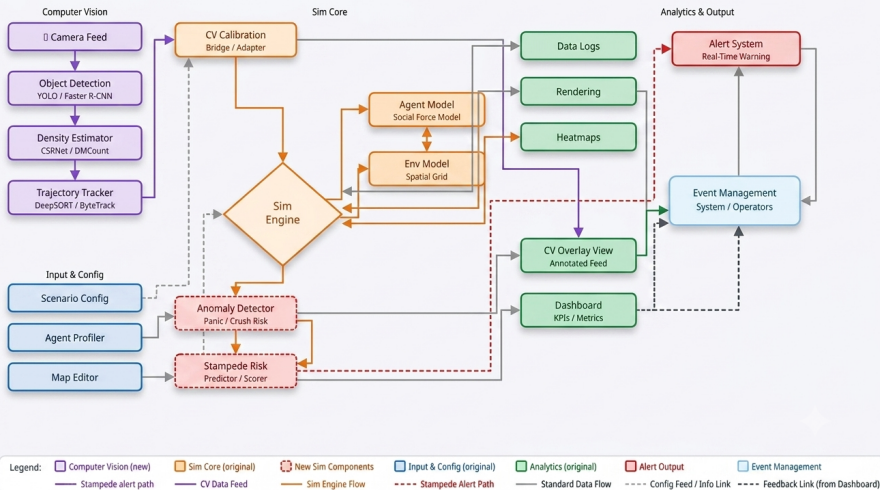
- **Python:** Computer vision and backend logic
- **FastAPI:** API and live video streaming
- **OpenCV:** Camera access and frame annotation
- **Ultralytics YOLO:** Person detection
- **React + Vite:** Browser-based dashboard

Planned Extensions

- **DeepSORT / ByteTrack:** Multi-person tracking
- **CSRNet:** Density estimation for high-density scenes
- **Mesa / NumPy:** Agent-based simulation
- **Plotly / Matplotlib:** Heatmaps and graphs
- **A*:** Dynamic rerouting to safer exits

System Design / Architecture

Architecture — CV-Integrated Agent-Based Crowd Simulation



Implemented Prototype Architecture

Backend

- FastAPI server
- OpenCV video capture
- YOLO person detection
- Annotated MJPEG stream
- People count and risk statistics API

Frontend

- React dashboard
- Camera/video source input
- Confidence slider
- Density-zone toggle
- Live annotated video output

Camera / Video Feed → FastAPI + OpenCV → YOLO Detection → Annotated Stream → React Dashboard

Current Demo Implemented

A working web prototype has been developed using a FastAPI backend and React frontend for live person detection and annotated video output.

- Live webcam/video source support.
- YOLO-based person detection.
- Bounding boxes drawn around detected people.
- Real-time people count and FPS display.
- 3x3 density-zone overlay for basic spatial crowd analysis.
- Low / Moderate / High crowd-risk label based on detected count.
- Browser dashboard with source input, confidence control, and start/stop interaction.

Challenges & Roadblocks

Technical Challenges

- Real-time inference needs sufficient CPU/GPU performance.
- Detection accuracy drops under poor lighting, blur, and low camera angles.
- Dense crowds create occlusion, causing missed detections.
- Live camera permissions and dependency setup vary by system.

Research Challenges

- Real stampede datasets are limited and sensitive.
- Camera pixels must be calibrated to real-world distance for ped/m^2 density.
- Panic behavior and pushing forces are difficult to model reliably.
- False alarms must be minimized for practical event use.

Project Timeline / Status

Phase	Work	Status
Phase 1	Literature review, problem study, and architecture design	Completed
Phase 2	FastAPI backend and React frontend setup	Completed
Phase 3	Live YOLO person detection with annotated browser stream	Completed
Phase 4	Zone-wise density overlay and basic risk indicator	Completed
Phase 5	Multi-object tracking, speed, and direction estimation	Next
Phase 6	Agent-based simulation and Social Force Model integration	Pending
Phase 7	Heatmaps, alerts, testing, and final evaluation	Pending

Current Status

The project has moved from literature review and architecture design to a working initial computer-vision prototype.

Remaining Work

- Integrate DeepSORT or ByteTrack for stable multi-person tracking.
- Estimate each detected person's speed, direction, and trajectory.
- Convert camera detections into real-world floor-plan coordinates.
- Improve density estimation from simple zone count to people per square meter.
- Build the agent-based simulation using Social Force Model movement rules.
- Add bottleneck detection, panic propagation, and crowd pressure index.
- Generate real-time heatmaps and operator-facing alert messages.
- Evaluate accuracy, FPS, latency, false-positive rate, and alert lead time.

Conclusion

The integrated computer vision and agent-based crowd simulation framework successfully establishes a proactive safety tool for dynamic crowd control:

- **Real-Time Synchronization:** Combines live telemetry with microscopic agent trajectories to model emergent behaviors without divergence.
- **Proactive Warning Alerts:** Accurately predicts dynamic bottlenecks and safety-critical pressures up to 30 seconds before a crush occurs.
- **Strategic Interventions:** Provides dynamic routing guidance (A^*) to mitigate potential crowd emergencies.
- **Safe Infrastructure Planning:** Offers event organizers and managers a virtual environment to safely optimize space layout and evacuation protocols.

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